

Problem 1.4

The simplest example of interest is a loan agreement two children might make: “I will lend you a dollar, but every day you keep it, you owe me one more penny.” Write down a formula expressing the amount value after t days.

Solution

The initial amount is \$1, and each day adds \$0.01 to the amount owed. Therefore, after t days, the amount value is given by the linear formula

$$f(t) = 1 + 0.01t.$$

Hence, the required formula is

$$\boxed{f(t) = 1 + 0.01t}.$$